

Sampark Bhol

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SUMMARY

Aspiring AI and Machine Learning engineer driven by an obsessive curiosity for building production grade intelligent systems end to end across natural language processing, computer vision, predictive analytics, and quantum machine learning. Currently a Founding Machine Learning Engineer at CLR3 Ventures, architecting autonomous LLM agent pipelines, transformer similarity engines, and full ML observability stacks from the ground up. Published Elsevier researcher and active open source contributor who reads research papers cover to cover, fine tunes frontier models for fun, and ships code that lands in production. Constantly eager to master new frameworks, dive into emerging compute paradigms, and translate cutting research into systems that matter.

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering

Vellore, India

2022 to 2026

- Relevant Coursework: Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networking, Object Oriented Programming, Statistics, Probability, Linear Algebra.

WORK EXPERIENCE

Founding Machine Learning Engineer (Intern)

CLR3 Ventures (Spizen Technologies)

Jan. 2026 to Present

Bengaluru, India

- Built a profitability prediction system with ensemble gradient boosting on streaming Kafka feature pipelines, hitting 92% accuracy on revenue forecasts and powering dynamic pricing across business verticals.
- Shipped a transformer embedding similarity engine with approximate nearest neighbor search on ClickHouse and a neural collaborative filtering recommender serving 10K+ queries per second in production.
- Architected an autonomous long form content and synthetic data generation pipeline orchestrating multi stage LLM agents through planning, drafting, summarization, and validation, with retrieval based deduplication and policy aware filtering for downstream training corpora.
- Established the founding ML infrastructure on Apache Kafka for event driven streaming with Grafana dashboards for system observability, model drift detection, and inference latency monitoring, plus CI/CD pipelines with A/B testing for safe model rollouts.

Quantum Computing and Research Intern

QuantumX (Appetite Studio)

Dec. 2025 to Jan. 2026

Remote

- Researched and developed hybrid quantum classical machine learning models combining variational quantum circuits with classical neural networks for combinatorial optimization, exploring quantum advantage in feature space mapping and kernel methods.
- Designed quantum optimization protocols using QAOA and VQE algorithms, and investigated quantum safe cryptographic architectures and post quantum security frameworks for AI model integrity in distributed systems.
- Authored the open source Quark project, a machine learning library that learns semantic similarity embeddings for quantum circuits with a CLI deduplication tool over QASM repositories, and first authored a research journal on quantum enhanced machine learning for resource allocation and sustainability optimization, currently under review for publication.

AI and ML Engineer Intern

CSM Technologies

May 2025 to July 2025

Odisha, India

- Shipped production ready Retrieval Augmented Generation pipelines with LangChain, LangFlow, ChromaDB, and Pinecone vector stores, lifting domain specific accuracy by 25% via tuned chunking strategies, prompt engineering iterations, and retrieval pipeline tuning.
- Built BERT and GPT class transformer applications for text classification, named entity recognition, and document summarization, with end to end generative AI solutions, real time inference, and vector database integration.
- Fine tuned large language models using LoRA and QLoRA parameter efficient methods, cutting training compute by 60% while preserving task performance parity with full fine tuning baselines, then containerized the stack with Docker and orchestrated multi service deployments on Kubernetes.

Deep Learning and Software Development Intern

National Informatics Centre, MeitY, Government of India

May 2024 to July 2024

Odisha, India

- Built anomaly detection algorithms with Isolation Forest, DBSCAN, Z score statistics, and graph based analysis on 1M+ record government datasets, reaching 95% precision recall on irregular pattern identification.
- Developed predictive models with Logistic Regression, SVM, Gradient Boosting, K Means, Hierarchical Clustering, and PCA, paired with comprehensive exploratory data analysis and feature engineering on structured and unstructured data.
- Implemented automated ETL pipelines and data warehousing in Python and SQL with schema validation and quality profiling, cutting processing time by 70% and shipping Matplotlib, Seaborn, and Plotly dashboards adopted across multiple government departments.

PROJECTS

- Cairn** | *Python, msgpack RPC, SQLite, Git Worktrees, LLM Agents* 2026
- Built an autonomous experiment runner letting LLM coding agents iteratively improve a codebase while keeping a persistent wiki of every attempted experiment, addressing the memory and metric drift limitations of prior agent loops.
 - Architected as a Python daemon and thin client over msgpack framed RPC with SQLite WAL state store, git worktree isolation per run, and shell injection safe subprocess execution with process group isolation for graceful shutdown.
 - Implemented hard budget caps on experiments, wallclock, and cost, plus an anti Goodhart holdout evaluator running in fresh worktrees every Nth experiment, with comprehensive observability via per experiment logs and 67+ regression tests.
- Browser-LLM** | *JavaScript, WebGPU, WebAssembly, Transformers.js* 2025
- Engineered a fully client side AI chatbot with WebGPU hardware acceleration delivering sub second token generation latency and zero backend dependency, ensuring complete data privacy through on device inference.
 - Implemented model loading optimizations with lazy weight streaming, KV cache management, and memory efficient attention mechanisms to run quantized GGUF and ONNX models within browser memory constraints on consumer hardware.
 - Built a progressive web app with conversation history persistence, model switching capabilities, and adaptive inference settings for balancing response quality versus generation speed based on device capabilities.
- DataWiz-Pro** | *Python, React, Pandas, Scikit-learn, Plotly, FastAPI* 2025
- Developed an AI powered AutoML analytics platform with automated hyperparameter tuning via Bayesian optimization, automated feature selection, and model selection across regression, classification, and clustering tasks.
 - Engineered end to end data pipelines supporting CSV, JSON, and SQL ingestion with automated preprocessing, missing value imputation, outlier detection, and feature scaling for production ready analytics workflows.
 - Integrated advanced interactive data visualization and business intelligence dashboards with real time data exploration, statistical profiling, and exportable reports for non technical stakeholders to perform data driven decision making.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL, R, Golang, Solidity, HTML5, CSS3

AI/ML and Deep Learning: TensorFlow, PyTorch, Scikit-learn, Keras, XGBoost, LightGBM, CatBoost, Hugging Face Transformers, LangChain, LangGraph, LangFlow, Haystack, OpenCV, MediaPipe, FAISS, Neural Networks, GANs, Variational Autoencoders, Attention Mechanisms, Reinforcement Learning, Transfer Learning, Few Shot Learning

NLP and Generative AI: BERT, GPT, Sentence Transformers, RAG Pipelines, Prompt Engineering, Fine Tuning (LoRA, QLoRA, PEFT), Text Classification, Named Entity Recognition, Semantic Search, Embeddings, Tokenization, SpaCy, NLTK

Data Science and Analytics: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly, Statistical Hypothesis Testing, A/B Testing, Bayesian Inference, Feature Engineering, Exploratory Data Analysis, Time Series Analysis, PowerBI, Tableau

MLOps and Infrastructure: Docker, Kubernetes, Rancher, MLflow, Kubeflow, Weights and Biases, TorchServe, TF Serving, Horovod, DeepSpeed, Jenkins, GitLab CI, GitHub Actions, Model Monitoring, Experiment Tracking, DVC, Apache Spark (PySpark), Kafka, Redpanda, Hadoop, Apache Airflow, ETL/ELT Pipelines, Delta Lake

Databases and Cloud: PostgreSQL, MySQL, MongoDB, Redis, ClickHouse, Pinecone, Weaviate, ChromaDB, Firebase, SQLite, DynamoDB, Elasticsearch, AWS (Lambda, S3, EC2, SageMaker, Bedrock), GCP (Vertex AI), Git, Terraform

Web and Tools: React, Node.js, Flask, Django, FastAPI, REST APIs, GraphQL, Three.js, Flutter, Vite, WebGPU, WebAssembly, Figma, Postman, n8n, Ollama, Jupyter, Streamlit, Gradio, MCP

ACHIEVEMENTS AND CERTIFICATIONS

Research: “Energy Efficient Hybrid WSN Protocol” in Elsevier Procedia Computer Science (2025), proposing a novel energy efficient hybrid routing protocol for wireless sensor networks with significant efficiency improvements for large scale IoT deployments through optimized cluster head selection and data aggregation strategies.

Open Source: Authored Quark, a quantum circuit semantic similarity embedding library with a QASM deduplication CLI; active contributor to community driven quantum computing initiatives and AI ML research projects on GitHub.

Leadership: AIML Lead at IEEE Computer Society VIT, organizing hackathons with 200+ participants and conducting technical workshops on deep learning, NLP, and computer vision for the student community.

Certifications: Oracle Generative AI Cloud Infrastructure Professional, Oracle Cloud Infrastructure Associate, IBM Deep Learning Fundamentals, NVIDIA Building Transformer Based NLP Applications, ISAA Saviynt Identity Security.